SYNTHETIC CATEGORY THEORY

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FUNCTOR CATEGORIES

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Recall: MAPPING ANIMAE AXIOM

C, D categories, there is an ANIMA $\mathrm{Map}(C, D)$

with $ev_{\mathrm{Map}} : \mathrm{Map}(C, D) \times C \longrightarrow D$

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If X ANIMA and $h : X \times C \to D$, there is (CURRYING) $h_c : X \to \mathrm{Map}(C,D)$ equipped with $(h_c)^{\mu} = h$ where

(UNCURRYING) $a^{\mu} := ev_{\mathrm{Map}} \circ (a \times id_c) : X \times C \to D$ for $a : X \to \mathrm{Map}(C,D)$

For $a, b : X \to \mathrm{Map}(C,D)$ the induced functor

$(-)^{\mu} : (a = b) \longrightarrow (a^{\mu} = b^{\mu})$ is an EQUIVALENCE

Def (FUNCTOR CATEGORY)

A FUNCTOR CATEGORY from C to D is a category F EQUIPPED with $e : F \times C \longrightarrow D$ s.t. for any T

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$$u_T : \mathrm{Map}(T, F) \xrightarrow{\sim} \mathrm{Map}(T \times C, D)$$

where u_T is the CURRYING of the composite

$$\mathrm{Map}(T,F) \times T \times C \xrightarrow{ev_{\mathrm{Map}} \times id_C} F \times C \xrightarrow{e} D$$

and for $g : T \to F$ $\quad u_T(g) = e \circ (g \times id_C)$

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Prop (UNIQUENESS)

If F and F' are both FUNCTOR CATEGORIES from C to D with e and e' respectively, then THERE IS AN EQUIVALENCE $\varphi : F \xrightarrow{\sim} F'$ and IDENTIFICATION $\sigma : e \circ (\varphi \times id_C) = e'$

AXIOM B.8 (FUNCTOR CATEGORY)

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T category, V_T inverse of $\mathcal{U}_T$, $f: T \times C \longrightarrow D$ define the CURRYING OF f as

$$f_c := V_T(f): T \longrightarrow \mathrm{Fun}(C,D)$$

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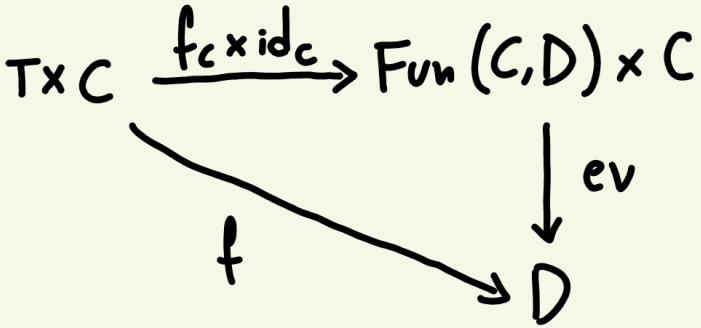
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$$g^{\mu} := T \times C \xrightarrow{g \times id_C} \mathrm{Fun}(C,D) \times C \xrightarrow{ev} D$$

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NOTE: • $g^{\mu} \equiv \mu_T(g)$

• $(g = h) \longrightarrow (g^{\mu} = h^{\mu})$

$\beta : g = h \qquad id_{id_C} : id_C = id_C$

$\Rightarrow \beta \times id_{id_C} : g \times id_C = h \times id_C$

and POSTWISKERING WITH ev

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Consider $g, h : * \longrightarrow \mathrm{Map}(T, \mathrm{Fun}(C,D))$ and

WHISKERING WITH AN EQUIV INDUCES EQUIV OF IDENTIFICATIONS

REMARK Take $T = *$ the ABSOLUTE OBJECTS $* \longrightarrow \mathrm{Fun}(C,D)$ correspond to FUNCTORS $C \simeq * \times C \longrightarrow D$

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EVALUATION AT A POINT

C, D categories, $x : * \longrightarrow C$ we define

$$ev_x : \mathrm{Fun}(C,D) \simeq \mathrm{Fun}(C,D) \times * \xrightarrow{id \times x} \mathrm{Fun}(C,D) \times C \xrightarrow{ev} D$$

FUNCTORIALITY

Let $g : D \longrightarrow D'$ we define $(g \circ -) : \mathrm{Fun}(C,D) \longrightarrow \mathrm{Fun}(C,D')$

as the CURRYING of

$$\mathrm{Fun}(C,D) \times C \xrightarrow{ev} D \xrightarrow{g} D'$$

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that comes equipped with

$$\begin{array}{ccc} \mathrm{Fun}(C,D) \times C & \xrightarrow{ev} & D \\ {\scriptstyle (g\circ -)\times id_C}\downarrow & & \downarrow{\scriptstyle g} \\ \mathrm{Fun}(C,D') \times C & \xrightarrow[ev]{} & D' \end{array}$$

FUNCTORIALITY

Let $g : D \longrightarrow D'$ we define $(g \circ -) : Fun(C,D) \longrightarrow Fun(C,D')$ as the currying of

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For $f : C' \longrightarrow C$, $(- \circ f) : Fun(C,D) \longrightarrow Fun(C',D)$ is defined as the currying of

$$Fun(C,D) \times C' \xrightarrow{id \times f} Fun(C,D) \times C \xrightarrow{ev} D$$

equipped with

$$\begin{array}{ccc} Fun(C,D) \times C' & \xrightarrow{id \times f} & Fun(C,D) \times C \\ {\scriptstyle (- \circ f) \times id_{C'}} \downarrow & & \downarrow {\scriptstyle ev} \\ Fun(C',D) \times C' & \xrightarrow[ev]{} & D \end{array}$$

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Similarly identification $\alpha : f = f'$ and $\beta : g = g'$ induce identifications $(- \circ \alpha) : (- \circ f) = (- \circ f')$

$$(\beta \circ -) : (g \circ -) = (g \circ -)$$

PROPERTIES OF FUNCTOR CATEGORIES

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Recall For $f: C \to D$ TFAE

- f is an EQUIVALENCE
- for any E, $f_*: \mathrm{Map}(E, C) \to \mathrm{Map}(E, D)$ is an EQUIV.
- for any E, $f^*: \mathrm{Map}(D, E) \to \mathrm{Map}(C, E)$ is an EQUIV.

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(I) For C category, $\mathrm{Fun}(C,*)$ IS CONTRACTIBLE

$$\mathrm{Map}(T, \mathrm{Fun}(C,*)) \simeq \mathrm{Map}(T \times C, *) \simeq * \simeq \mathrm{Map}(T, *)$$

(II) C, D, E categories, the following is an EQUIVALENCE

$$(pr_1 \circ -, pr_2 \circ -) : Fun(C, D \times E) \longrightarrow Fun(C, D) \times Fun(C, E)$$

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$$\begin{aligned} Map(T, Fun(C, D \times E)) &\simeq Map(T \times C, D \times E) \\ &\simeq Map(T \times C, D) \times Map(T \times C, E) \\ &\simeq Map(T, Fun(C, D)) \times Map(T, Fun(C, E)) \\ &\simeq Map(T, Fun(C, D) \times Fun(C, E)) \end{aligned}$$

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(III) T category, $f : C \to E$, $g : D \to E$, the following is an EQUIVALENCE

$$(pr_1 \circ -, pr_2 \circ -) : Fun(T, C \times_E D) \longrightarrow Fun(T, C) \times_{Fun(T,E)} Fun(T, C)$$

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Show that under $Map(S, Fun(T, -))$ the SQUARE REMAINS A PULLBACK, and AFTER UNCURRYING show that

$$\begin{array}{ccc} Map(S \times T, C \times_E D) & \longrightarrow & Map(S \times T, D) \\ \downarrow & & \downarrow \\ Map(S \times T, C) & \longrightarrow & Map(S \times T, E) \end{array}$$

IS THE SAME AS APPLYING $Map(S \times T, -)$ TO THE ORIGINAL PULLBACK

(Ⅳ) For C, D, E there is a preferred EQUIVALENCE

$$\mathrm{Fun}\left(C, \mathrm{Fun}(D, E)\right) \xrightarrow{\sim} \mathrm{Fun}\left(C \times D, E\right)$$

(IV) For C, D, E there is a preferred EQUIVALENCE

$$\mathrm{Fun}(C, \mathrm{Fun}(D,E)) \xrightarrow{\sim} \mathrm{Fun}(C \times D, E)$$

CURRYING the composite

$$\mathrm{Fun}(C,\mathrm{Fun}(D,E)) \times C \times D \xrightarrow{ev \times id_D} \mathrm{Fun}(D,E) \times D \xrightarrow{ev} D$$

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$$\mathrm{Fun}(C, \mathrm{Fun}(D,E)) \xrightarrow{\sim} \mathrm{Fun}(C \times D, E)$$

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(V) For C category, $ev_* : \mathrm{Fun}(*, C) \xrightarrow{\sim} C$ is an EQUIV.

$$\mathrm{Map}(E, \mathrm{Fun}(*,C)) \xrightarrow[\sim]{(-)^{\mu}} \mathrm{Map}(E \times *, C) \xrightarrow[\sim]{(i_E)^*} \mathrm{Map}(E, C)$$

$$ev_* \circ -$$

where $i_E : E \times * \xrightarrow{\sim} E$

(VI) For C category, $\mathrm{Fun}(\emptyset, C)$ is CONTRACTIBLE

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(VII) For C, D, E the following is an EQUIVALENCE

$$(i_C^*, i_D^*) : \mathrm{Fun}(C \amalg D, E) \xrightarrow{\sim} \mathrm{Fun}(C, E) \times \mathrm{Fun}(D, E)$$

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(VIII) For C, D categories there is a preferred EQUIVALENCE

$$\mathrm{Map}(C, D) \xrightarrow{\sim} \mathrm{Fun}(C, D)^{\simeq}$$

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(VIII) For C, D categories there is a preferred EQUIVALENCE

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Consider $(ev_{\mathrm{Map}})_C : \mathrm{Map}(C, D) \longrightarrow \mathrm{Fun}(C, D)$

$$\mathrm{Fun}(C, D)^{\simeq} \simeq \mathrm{Map}(*, \mathrm{Fun}(C, D)) \simeq \mathrm{Map}(* \times C, D) \simeq \mathrm{Map}(C, D)$$

PUSHOUT SQUARES

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Def A commutative square of categories

$$\begin{array}{ccc} C' & \xrightarrow{u} & C \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ D' & \xrightarrow[v]{} & D \end{array}$$

is called a **PUSHOUT SQUARE** if for any E the following is a PULLBACK SQUARE

$$\begin{array}{ccc} \mathrm{Map}(D,E) & \xrightarrow{v^*} & \mathrm{Map}(D',E) \\ {\scriptstyle f^*}\downarrow & & \downarrow{\scriptstyle (f')^*} \\ \mathrm{Map}(C,E) & \xrightarrow[u^*]{} & \mathrm{Map}(C',E) \end{array}$$

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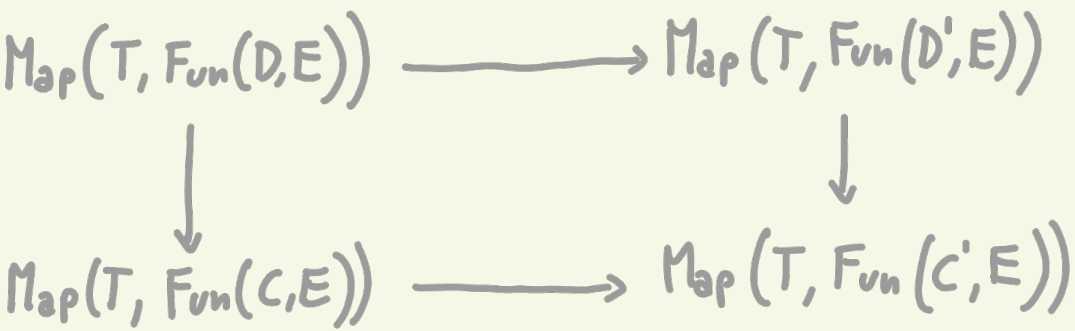
Prop A commutative square

$$\begin{array}{ccc} C' & \xrightarrow{u} & C \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ D' & \xrightarrow[v]{} & D \end{array}$$

IS A PUSHOUT if and only if for any E the square

$$\begin{array}{ccc} Fun(D,E) & \xrightarrow{v^*} & Fun(D',E) \\ {\scriptstyle f^*}\downarrow & & \downarrow{\scriptstyle (f')^*} \\ Fun(C,E) & \xrightarrow{u^*} & Fun(C',E) \end{array}$$

IS A PULLBACK


Map(T, Fun(D,E))
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$$\begin{array}{ccc} \mathrm{Map}(T, \mathrm{Fun}(D,E)) & \longrightarrow & \mathrm{Map}(T, \mathrm{Fun}(D',E)) \\ \downarrow & & \downarrow \\ \mathrm{Map}(T, \mathrm{Fun}(C,E)) & \longrightarrow & \mathrm{Map}(T, \mathrm{Fun}(C',E)) \end{array}$$

is equiv to

$$\begin{array}{ccc} \mathrm{Map}(T \times D, E) & \longrightarrow & \mathrm{Map}(T \times D', E) \\ \downarrow & & \downarrow \\ \mathrm{Map}(T \times C, E) & \longrightarrow & \mathrm{Map}(T \times C', E) \end{array}$$

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in equiv to

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in equiv to

$$\begin{array}{ccc} \mathrm{Map}(D, \mathrm{Fun}(T,E)) & \longrightarrow & \mathrm{Map}(D', \mathrm{Fun}(T,E)) \\ \downarrow & & \downarrow \\ \mathrm{Map}(C, \mathrm{Fun}(T,E)) & \longrightarrow & \mathrm{Map}(C', \mathrm{Fun}(T,E)) \end{array}$$

$$\begin{array}{ccc} \mathrm{Map}(T, \mathrm{Fun}(D,E)) & \longrightarrow & \mathrm{Map}(T, \mathrm{Fun}(D',E)) \\ \downarrow & & \downarrow \\ \mathrm{Map}(T, \mathrm{Fun}(C,E)) & \longrightarrow & \mathrm{Map}(T, \mathrm{Fun}(C',E)) \end{array}$$

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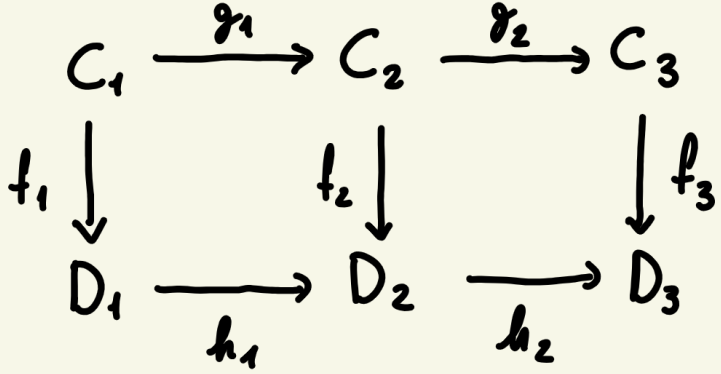
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is equiv to

$$\begin{array}{ccc} \mathrm{Map}(D, E) & \longrightarrow & \mathrm{Map}(D', E) \\ \downarrow & & \downarrow \\ \mathrm{Map}(C, E) & \longrightarrow & \mathrm{Map}(C', E) \end{array}$$

Take $T = *$

Prop (PASTING LEMMA) For the commutative digram



If the LEFT SQUARE IS A PUSHOUT, then the RIGHT SQUARE IS A PUSHOUT if and only if the OUTER SQUARE IS A PUSHOUT